

Homework 5: Multiple Linear Regression

Math 141, Week 5

Your name here

Due: Thursday, October 8th at 9:00am on Gradescope as a PDF

In this homework assignment you will:

- Practice interpreting regressions with categorical predictors
- Practice using visualizations to perform exploratory data analysis and assess linearity before building a regression model
- Practice applying transformations to help satisfy LINE assumptions
- Build and interpret linear regression models with interactions

Add any collaborators here!

I collaborated with...

Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this class. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

```
#load the necessary packages
library(tidyverse)
library(ggml)
library(moderndiver)
library(pdxTrees)
```

Exercises

Exercise 1: Final Project - Dataset & Research Questions

As a reminder, throughout the semester you will work in your project group to answer a research question using real-world data and techniques covered in class, culminating in a technical report and presentation.

a. Describe the dataset your group is considering for the final project. Where does it come from, and what does a row/observation represent?

b. List 2-3 draft research questions your group is considering.

c. Are there any concerns about this dataset (e.g., missing data, sample size, variables you wish you had) that might affect which research questions are feasible?

Exercise 2: Factors (Categorical Predictors)

We will again work with the `pdxTrees` data package. Run the following code chunk to obtain a data frame of the four parks closest to Reed, subsetted to only include trees of the **Genus** *Pseudotsuga*, *Acer*, *Quercus*, or *Cornus*. This data frame includes some additional modification to clean up the data. The chunk also shows how many trees of each **Genus** are in the `near_reed` dataset.

```

near_reed <- get_pdxTrees_parks() %>%
  filter(Park %in% c("Crystal Springs Rhododendron Garden",
                    "Kenilworth Park",
                    "Eastmoreland Garden",
                    "Berkeley Park")) %>%
  mutate(Type = case_when(
    Functional_Type == "BD" | Functional_Type == "CD" ~ "Deciduous",
    Functional_Type == "ED" | Functional_Type == "CE" ~ "Evergreen"
  )) %>%
  mutate(Crown_Width = (Crown_Width_NS + Crown_Width_EW)/2) %>%
  drop_na(Type) %>%
  filter(Genus %in% c("Pseudotsuga", "Acer", "Quercus", "Cornus"))

near_reed %>% group_by(Genus) %>% summarize(Count=n())

```

```

# A tibble: 4 x 2
  Genus      Count
  <chr>    <int>
1 Acer         104
2 Cornus        36
3 Pseudotsuga  193
4 Quercus       52

```

a. In the chunk below, construct a linear model to analyze the differences in the Carbon_Sequestration_lb for each level in Genus. **How many indicator variables are displayed in the regression table, and what are they? Which level of the Genus variable does not have an indicator variable?**

b. Write down the formula for the linear model for Carbon_Sequestration_lb as a function of Genus.

c. Compute the mean `Carbon_Sequestration_lb` for trees in each `Genus`. Then, identify where the value for the `Acer` genus appears in the regression table.

d. Demonstrate how the mean `Carbon_Sequestration_lb` for trees from the `Cornus`, `Pseudotsuga`, or `Quercus` genera can be calculated from the *regression table*.

e. By default, R chooses the alphabetically first level of the explanatory variable to serve as the baseline. However, the order can be changed in R using some extra code. In the following code chunk, we re-order the `Genus` variable so that the largest category, `Pseudotsuga`, is the *baseline* group.

```
near_reed <- near_reed %>%  
  mutate(Genus = fct_relevel(Genus, "Pseudotsuga", "Acer", "Quercus", "Cornus"))
```

Create a new linear model for `Carbon_Sequestration_lb` as a function of `Genus`. Comment on how the coefficients estimates have changed.

f. Suppose a particular tree is a Bigleaf Maple, which is in the `Acer` genus. **What is the predicted Carbon Sequestration using the model constructed in part (a)? What is the predicted Carbon Sequestration using the model constructed in part (e)? How do the predictions compare?**

Exercise 3: Regression with Transformation

Problems 3 & 4 use an extended version of the [Opportunity Insights dataset](#) (introduced in the activity from the *Data Wrangling* lecture) that has been joined with a dataset from The National Association of College and University Business Officers containing information on university endowment assets from 2016.¹ The data are in the file `colleges_endow.Rdata`; for simplicity, the data frame is named `colleges`. The following table provides a list of the variables in the dataset and their descriptions. (Note: It is easier to read the table in the Rendered PDF document.)

Variable	Description
<code>name</code>	name of college
<code>region</code>	region, 1 for Northeast, 2 for Midwest, 3 for South, and 4 for West
<code>state</code>	state
<code>tier_name</code>	tier defined by selectivity and type
<code>type</code>	coded 1 for public, 2 for private non-profit, 3 for for-profit
<code>exp_instr_pc_2013</code>	instructional expenditures per student in 2013
<code>ipeds_enrollment_2013</code>	total undergraduate enrollment in Fall 2013
<code>sticker_price_2013</code>	average annual cost of attendance in 2013
<code>scorecard_netprice_2013</code>	net annual cost of attendance for bottom income quintile in 2013
<code>grad_rate_150_p_2013</code>	percentage of students graduating within 150% of normal time in 2013
<code>avgfacsal_2013</code>	average faculty salary in 2013
<code>sat_avg_2013</code>	average SAT scores (scaled to 1600) in 2013
<code>endowment_pc_2000</code>	endowment assets per student in 2000
<code>mr_kq5_pq1</code>	mobility rate, top 20% of the income distribution
<code>mr_ktop1_pq1</code>	mobility rate, top 1% of the income distribution
<code>par_median</code>	median parent household income
<code>par_q1</code>	fraction of parents in first (bottom) income quintile
<code>par_q2</code>	fraction of parents in second income quintile
<code>par_q3</code>	fraction of parents in third income quintile
<code>par_q4</code>	fraction of parents in fourth income quintile
<code>par_q5</code>	fraction of parents in fifth (top) income quintile
<code>par_top5pc</code>	fraction of parents in top 5% of income distribution
<code>par_top1pc</code>	fraction of parents in top 1% of income distribution
<code>k_median</code>	median child individual earnings in 2014 (at age 34)
<code>k_top5pc</code>	fraction of children in top 5% of income distribution
<code>k_top1pc</code>	fraction of children in top 1% of income distribution
<code>fy2016_endowment_pc</code>	endowment assests per student in 2016
<code>fy2016_endowment_thousands</code>	total value of endowment in thousands USD in 2016

¹Specifically, the endowment data are from fiscal year 2016, which began on June 1, 2015 and ended on May 31, 2016. For the purposes of Math 141, it is fine to simply refer to these data as being collected in 2016.

Run the following R chunk to load the `colleges` data frame.

```
load("data/colleges_endow.Rdata")
```

One metric of college access is net tuition price for low- and middle-income students after accounting for financial aid. The `scorecard_netprice_2013` variable contains data on the net annual cost of attendance for students with parental household income in the bottom quintile (in 2013); these data come from the U.S. Department of Education's College Scorecard.

A university's endowment refers to the financial assets invested by a university to support its educational and research mission; individuals and organizations can donate to contribute to a university's endowment. The `fy2016_endowment_pc` variable contains data on the monetary value of endowment assets per student (in 2016), measured in USD.

Let's investigate the association between college access and endowment assets by building a model for college access using endowment assets.

a. First, explore both the net tuition price variable and endowment assets variable individually. **Describe the distribution of each variable in the context of the data, referencing numerical and graphical summaries as needed.**

b. Construct a plot showing the association between net tuition price and endowment assets. Be sure to map the response variable to the y -axis and the explanatory variable to the x -axis.

c. Describe what the plot in part (b) shows; i.e., based on the plot, **does there seem to be a relationship between college access and endowment assets? Does a line seem like a reasonable summary for the relationship between these variables?** Explain your reasoning.

d. Add a variable to the `colleges` dataset that contains the log-transformed version of `fy2016_endowment_pc` and name it `log_endow_pc`. That is, if `fy2016_endowment_pc` equals \$1000, then the log-transformed version equals $\log(1000) = \ln(1000) = 6.908$. Statisticians tend to refer to natural log as just “log”. The function in R to compute the (natural) log is `log()`.

e. Recreate the plot from part (b), but use the log-transformed version of `fy2016_endowment_pc` rather than the original variable.

f. Based on the plot in part (e), does a line seem like a reasonable summary of the relationship between net tuition price and log-transformed endowment assets? If so, calculate the correlation coefficient and interpret the sign and magnitude.

g. Construct a simple linear regression model for net tuition price using the log-transformed endowment assets variable. **Print out the model coefficients table.**

(Hint: Recall that you have the demo files from last lab with example code for building a linear regression model.)

h. Check the model diagnostics for the model from part (g), and **comment on them.**

(Hint: Recall that you have the demo files from last lab with example code for creating model diagnostic plots.)

Exercise 4: Regression Models with Equal vs Varying Slopes

In this problem, we will investigate whether the relationship between net tuition price for low- and middle-income students and endowment value seems to differ for Ivy Plus schools versus other schools.

a. Create a new variable that categorizes whether or not a school is Ivy Plus. (*Hint: Revisit the code in slide 25 from 2/4*)

b. Construct a plot showing the association between net tuition price and log-transformed endowment assets; color the points based on whether a school is in the Ivy Plus. **Describe what you see. How does the average net tuition price for low- and middle-income students at Ivy Plus schools compare to other schools?**

(*Hint: Revisit the code in slide 7 from 2/23 for creating the plot, and slide 30 on 2/4 for calculating group averages.*)

c. Construct a linear regression model for net tuition price using log-transformed endowment assets *and* whether a school is categorized as Ivy Plus. *Assume equal slopes in the model.* Print the model coefficients table **and interpret the coefficients in the context of the data.** **Be sure to comment on whether the intercept makes sense in this context.**

d. In part (b), you estimated the difference in average net tuition price for low- to middle-income students at schools in the Ivy Plus versus schools not in the Ivy Plus. **Explain the reasoning behind why this difference is not equal to the coefficient for the Ivy Plus predictor variable in the model from part (c).**

e. Construct a linear regression model for the net tuition price using log-transformed endowment assets and whether a school is categorized as Ivy Plus. *Assume varying slopes in the model.* Based on this model, **write out the formula for the regression line for just Ivy Plus schools and the formula for the regression line for just non-Ivy Plus schools.** Using these formulas, state **what is the estimated slope for schools in the Ivy Plus and what is the estimated slope for non-Ivy Plus schools.**

f. Produce a plot that illustrates the linear model fit in part (e). Based on this plot and your previous analyses, is there evidence suggesting that making a donation to a school's endowment increases access for low- and middle-income students to the same extent at Ivy Plus schools as non-Ivy Plus schools? **Explain your reasoning.**

(Hint: Revisit the code in slides 19-20 from 2/23)